

Distribution of Quadratic Forms

Recall,

Cochran's theorem

$Y \sim N(\underline{0}, I_n)$ and A is symmetric and idempotent.

Then $Y^T A Y \sim \chi_r^2$ where $r = \rho(A)$.

Result

$Y \sim N(\underline{0}, \Sigma)$, A is symmetric and $A\Sigma$ has eigenvalues 0 or 1.

Then $Y^T A Y \sim \chi_r^2$ where $r = \rho(A\Sigma)$.

Proof: $\Sigma = BB^T$ and $Y \stackrel{D}{=} BZ$ where $Z \sim N(\underline{0}, I)$.

$$Y^T A Y \stackrel{D}{=} Z^T B^T A B Z.$$

Note that the non-zero eigenvalues of $B^T A B$ is same as that of $AB B^T = A\Sigma$.

So, $B^T A B$ has the eigenvalue 1 with multiplicity $r = \rho(A\Sigma)$, and $(p - r)$ zero-eigenvalues.

This implies that the spectral decomposition of $B^T A B$ will be of the form:

$$B^T A B = U U^T \quad \text{with} \quad U^T U = I_r.$$

Thus,

$$Y^T A Y = Z^T B^T A B Z = Z^T U U^T Z = \|U^T Z\|^2.$$

$$Z \sim N(\underline{0}, I) \Rightarrow U^T Z \sim N(\underline{0}, I_r) \Rightarrow \|U^T Z\|^2 \sim \chi_r^2.$$

Result

$Y \sim N_p(\mu, \Sigma)$, $\Sigma > 0$. Then $(Y - \mu)^T \Sigma^{-1} (Y - \mu) \sim \chi_p^2$.

Craig's theorem

$Y \sim N_p(\underline{0}, I)$ and A, B are two symmetric and idempotent matrices.

Then $Y^T A Y \perp\!\!\!\perp Y^T B Y$ if $AB = 0$.

Proof: $Y_{\sim}^T A Y_{\sim} = \|A Y_{\sim}\|^2$ and $Y_{\sim}^T B Y_{\sim} = \|B Y_{\sim}\|^2$.

Enough to show that $A Y_{\sim} \perp\!\!\!\perp B Y_{\sim} \Leftrightarrow AB = 0$. [Property 6]

Example: $X \sim N_p(\mu, \sigma^2 I)$, $\sigma^2 > 0$.

$$S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (n-1)S_y^2 = Y_{\sim}^T \left(I - \frac{1}{n} \underline{1}\underline{1}^T \right) Y_{\sim}.$$

$$H_n = \left(I - \frac{1}{n} \underline{1}\underline{1}^T \right)$$

$$H_n \frac{1}{n} = \frac{1}{n} - \frac{1}{n} \underline{1}\underline{1}^T \cdot \frac{1}{n} \underline{1} = 0$$

Observe that $H_n^2 = H_n$.

Let $Z_{\sim} = \frac{1}{\sigma}(Y_{\sim} - \mu) \sim N(0, I_n)$, then $Z_{\sim}^T H_n Z_{\sim} \sim \chi_{n-1}^2$.

$$\frac{1}{\sigma^2} (Y_{\sim} - \mu)^T H_n (Y_{\sim} - \mu) \sim \chi_{n-1}^2.$$

Now,

$$(Y_{\sim} - \mu)^T H_n (Y_{\sim} - \mu) = Y_{\sim}^T H_n Y_{\sim} - 2\mu^T H_n Y_{\sim} + \mu^T H_n \mu.$$

If $H_n \mu = 0$ then the above equation is same as $Y_{\sim}^T H_n Y_{\sim}$.

$$(Y_{\sim} - \mu)^T H_n (Y_{\sim} - \mu) = Y_{\sim}^T H_n Y_{\sim}$$

If $\mu = c \underline{1}$ then

$$\frac{(n-1)S_Y^2}{\sigma^2} \sim \chi_{n-1}^2.$$

$$H_n = \left(I - \frac{1}{n} \underline{1}\underline{1}^T \right)$$

$$H_n \underline{1} = \underline{1} - \frac{1}{n} \underline{1}\underline{1}^T \underline{1} = 0.$$

Linear Regression

Recall the linear regression model with response y , and regressors/covariates $\{x_1, \dots, x_p\}$ and n obs. as follows:

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

where

$$\underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \underset{\sim}{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

We assume $E(\underset{\sim}{\varepsilon}) = 0$ and $\text{var}(\underset{\sim}{\varepsilon}) = \sigma^2 I$. Also, $\varepsilon_1, \dots, \varepsilon_n$ are independent.

Parameters: $(\underset{\sim}{\beta}, \sigma^2)$.

- $\underset{\sim}{\beta}$: vector of regression coefficients.
- σ^2 : error variance.

$$\left\{ \begin{array}{l} Y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \varepsilon_i, \quad i = 1, \dots, n \\ Y_i = \gamma_0 + \gamma_1 x_{1i} + \varepsilon_{1i} \\ Y_i = \gamma_0 + \gamma_2 x_{2i} + \varepsilon_{2i} \\ \vdots \\ Y_i = \gamma_0 + \gamma_p x_{pi} + \dots + \gamma_p x_{pi} + \varepsilon_{pi} \end{array} \right. \quad \begin{array}{l} \text{estimate} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \\ \text{estimate} \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_p \end{pmatrix} \end{array}$$

The above 2 estimates are not the same, they are modelling two different models hence the estimates won't be same.

Least Squares Method

We minimize

$$\underset{\sim}{\varepsilon}^T \underset{\sim}{\varepsilon} = (\underset{\sim}{Y} - X\underset{\sim}{\beta})^T (\underset{\sim}{Y} - X\underset{\sim}{\beta}) = \sum_{i=1}^n (Y_i - \beta_0 - \dots - \beta_p x_{pi})^2 = S(\underset{\sim}{\beta}),$$

w.r.t. $\underset{\sim}{\beta}$.

To minimize $S(\underset{\sim}{\beta})$ we differentiate $S(\underset{\sim}{\beta})$ w.r.t. $\underset{\sim}{\beta}$.

$$S(\underset{\sim}{\beta}) : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$$

$$\frac{\partial S(\beta)}{\partial \beta} = \frac{\partial}{\partial \beta} \left(\|Y\|^2 + \beta^T X^T X \beta - 2Y^T X \beta \right) = 0 + 2X^T X \beta - 2X^T Y.$$

First order condition:

$$\frac{\partial S(\beta)}{\partial \beta} = 0 \Rightarrow X^T X \beta = X^T Y.$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \text{when } (X^T X)^{-1} \text{ exists.}$$

Second order condition:

$$\frac{\partial^2 S(\beta)}{\partial \beta \partial \beta^T} = X^T X.$$

When $(X^T X)^{-1}$ exists, then $X^T X > 0$.

$\Rightarrow \hat{\beta}$ is the unique minima.

Useful derivatives, HW:

1. $\frac{\partial}{\partial x} (x^T A x)$ when A is symmetric.

$$= \frac{\partial}{\partial x_j} \left(\sum_{i,k} a_{ik} x_i x_k \right)_{j=1,\dots,p} = (A + A^T)x = 2Ax.$$

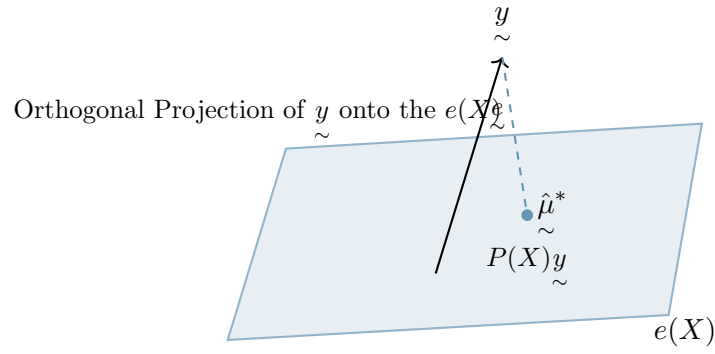
2. $\frac{\partial a^T x}{\partial x} = a.$

3. $\frac{\partial Ax}{\partial x} = A$ when A is symmetric.

Geometric interpretation

The problem is to find $\hat{\beta}$ s.t. $\|y - X\hat{\beta}\|^2$ is minimum.

i.e., to find the component $\hat{\mu}^*$ in the column space of X , $e(X)$, which is closest to y .



Orthogonal projection of y onto $e(X)$; $e(X)$ is the plane.

$$a_0 x_{\sim(0)} + a_1 x_{\sim(1)} + \cdots + a_p x_{\sim(p)} = X a_{\sim}$$

$$\hat{\mu}_{\sim}^* = P(X) y_{\sim} \quad \text{where } P(X) \text{ is the orthogonal projection matrix onto the } e(X).$$

Properties of Orthogonal projection matrix

- $P^2(X) = P(X) \Leftrightarrow P(X)[I - P(X)] = 0$ and also symmetric.
- If $\mu_{\sim} \in e(X)$ then $P(X)\mu_{\sim} = \mu_{\sim}$.
- $\rho(P(X)) = \rho(X) = \text{tr}(P(X))$.
- If $(X'X)^{-1}$ exists then $P(X) = X(X'X)^{-1}X'$.

If $(X'X)$ is singular then $P(X) = X(X'X)^{-}X'$, where A^{-} is any g-inverse of A .

Least Square solution (Geometrically)

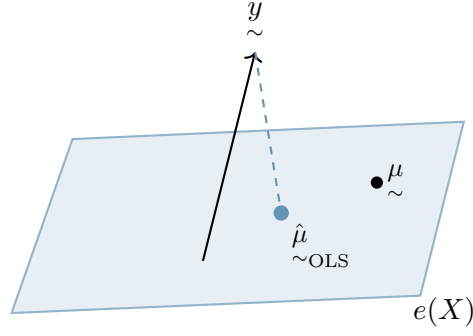
Let

$$\hat{\mu}_{\sim\text{OLS}} = X(X'X)^{-1}X'y_{\sim} \in e(X) = P(X)y_{\sim}$$

We will show that $\hat{\mu}_{\sim\text{OLS}}$ is the closest point of $y_{\sim}$ in $e(X)$.

Take any $\mu_{\sim} \in e(X)$

$$\begin{aligned} \|y_{\sim} - \mu_{\sim}\|^2 &= \|y_{\sim} - \hat{\mu}_{\sim\text{OLS}} + \hat{\mu}_{\sim\text{OLS}} - \mu_{\sim}\|^2 \\ &= \|y_{\sim} - \hat{\mu}_{\sim\text{OLS}}\|^2 + \|\hat{\mu}_{\sim\text{OLS}} - \mu_{\sim}\|^2. \end{aligned}$$



$\hat{\mu}_{\sim\text{OLS}}$ is the closest point of y in $e(X)$.

$$\begin{aligned}
 (y_{\sim} - \hat{\mu}_{\sim\text{OLS}})^T (\hat{\mu}_{\sim\text{OLS}} - \mu_{\sim}) &= (y_{\sim} - P(X)y_{\sim})^T (P(X)y_{\sim} - \mu_{\sim}) \\
 &= y_{\sim}^T (I - P(X)) (P(X)y_{\sim} - \mu_{\sim}) \\
 &= y_{\sim}^T \underbrace{(I - P(X))P(X)}_0 y_{\sim} - y_{\sim}^T \underbrace{(I - P(X))\mu_{\sim}}_{\mu_{\sim} - P(X)\mu_{\sim}} \\
 &= 0 \quad \text{on } \mu_{\sim} \in e(X).
 \end{aligned}$$

Thus, the distance is minimized if $\hat{\mu}_{\sim\text{OLS}} = \mu_{\sim}$.

So, the optimum point in $e(X)$ is:

$$\hat{\mu}_{\sim\text{OLS}} = P(X)y_{\sim}.$$

Q. For which $\hat{\beta}_{\sim}$, $P(X)y_{\sim} = X\hat{\beta}_{\sim}$?